LLM Decode Latency Breakdown

Standalone C TPOT decomposition by decode operator group under the default paper profile.

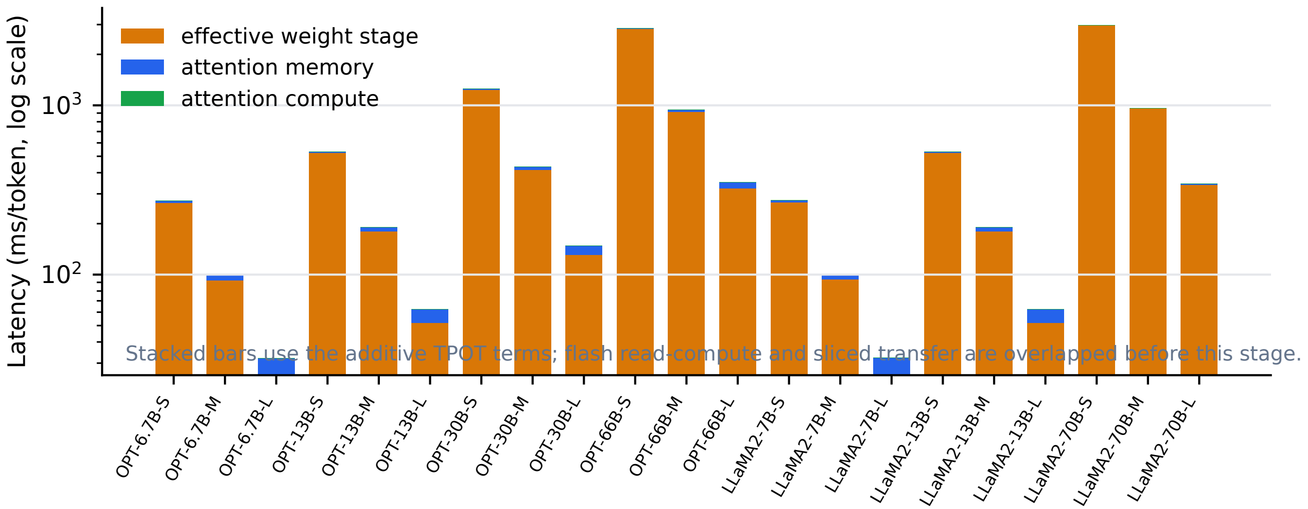
Fastest TPOT
32.139 ms/token
OPT-6.7B on Cambricon-LLM-L

LLaMA2-70B / L
344.473 ms/token
weight stage 98.6% of TPOT

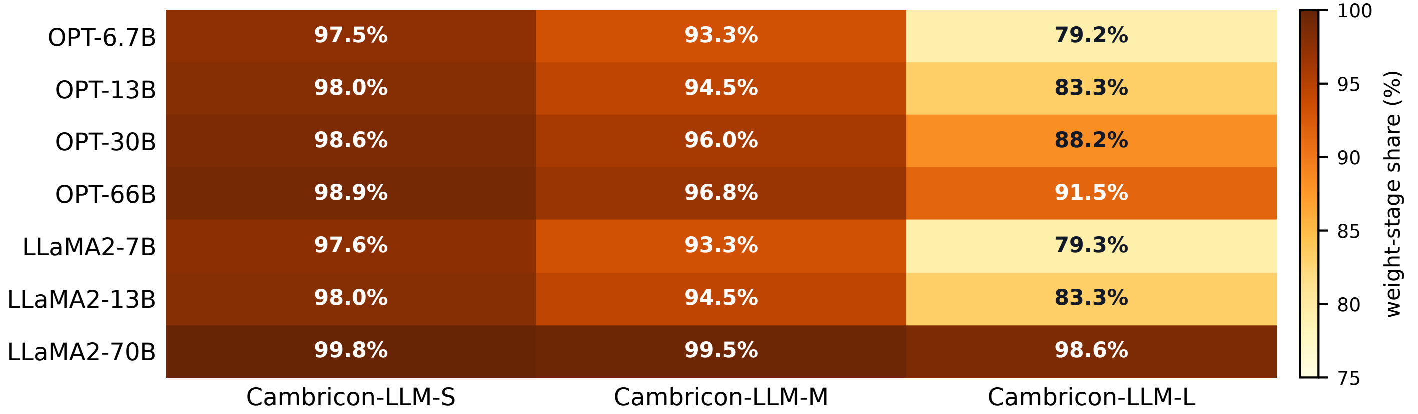
Attention tail / 70B-L
4.751 ms
memory 4.096 ms + compute 0.655 ms

TPOT formula
3 terms
weight stage + attention memory + compute

A. Absolute TPOT breakdown across all Figure 9 points



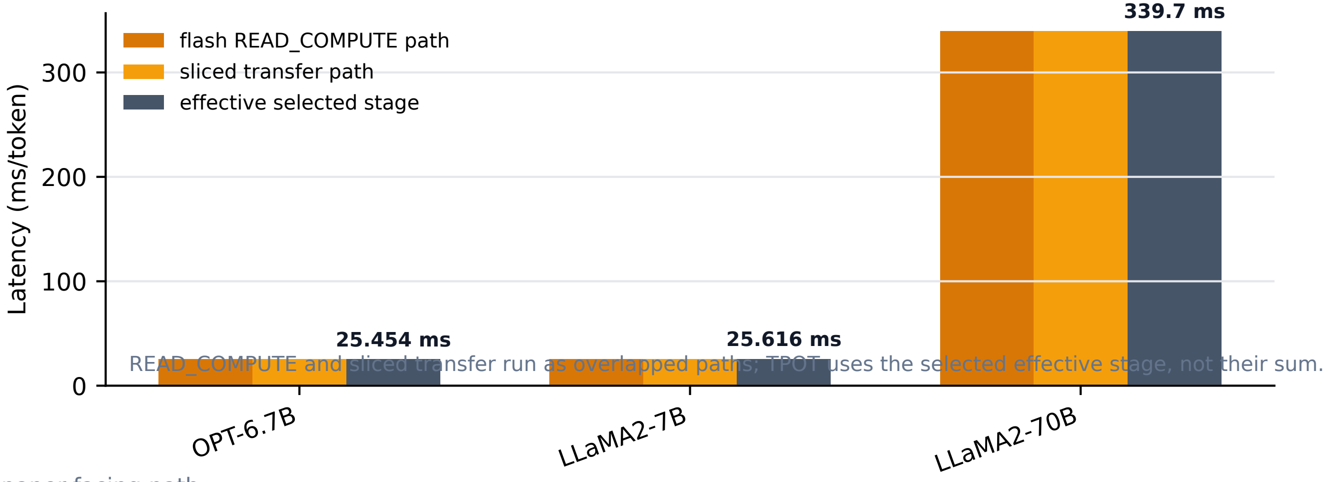
B. Weight-stage share of TPOT



C. Cambricon-LLM-L absolute operator latency table (ms/token)

	weight stage	attn memory	attn compute	TPOT
OPT-6.7B	25.454	6.554	0.131	32.139
OPT-13B	51.989	10.240	0.205	62.434
OPT-30B	131.0	17.203	0.344	148.6
OPT-66B	322.6	29.491	0.590	352.7
LLaMA2-7B	25.616	6.554	0.131	32.301
LLaMA2-13B	51.989	10.240	0.205	62.434
LLaMA2-70B	339.7	4.096	0.655	344.5

D. Overlapped flash-weight paths on Cambricon-LLM-L



Source: results/latency_breakdown.csv and results/npu_timing.csv. Breakdown is for decode TPOT under the standalone C paper-facing path.